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Concentrations of perfluoroalkyl substances (PFASs) in human embryonic and fetal organs from first, second, and third trimester pregnancies



Linn Salto Mamsen^{a,1}, Richelle D. Björvang^{b,c,*,1}, Daniel Mucs^{c,d}, Marie-Therese Vinnars^b, Nikos Papadogiannakis^e, Christian H. Lindh^f, Claus Yding Andersen^a, Pauliina Damdimopoulou^{b,c}

^a Laboratory of Reproductive Biology, Section 5712, The Juliane Marie Centre for Women, Children and Reproduction, University Hospital of Copenhagen, University of Copenhagen, Rigshospitalet, Blegdamsvej 9, 2100 Copenhagen, Denmark

^b Division of Obstetrics and Gynaecology, Department of Clinical Science, Intervention and Technology, K57 Karolinska University Hospital, Karolinska Institutet, 141 86 Stockholm, Sweden

^c Swetox, Karolinska Institute, Unit of Toxicology Sciences, Forskargatan 20, 151 36 Södertälje, Sweden

^d Unit of Work Environment Toxicology, Institute of Environmental Medicine, Box 210, Karolinska Institutet, 171 77 Stockholm, Sweden

^e Division of Pathology, Department of Laboratory Medicine, Karolinska Institutet, Karolinska University Hospital Huddinge H5, 141 83 Stockholm, Sweden

^f Division of Occupational and Environmental Medicine, Department of Laboratory Medicine, Medicon Village, Byggnad 402 A, Lund University, 223 61 Lund, Sweden

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ABSTRACT

Background: The persistent environmental contaminants perfluoroalkyl substances (PFASs) have gained attention due to their potential adverse health effects, in particular following early life exposure. Information on human fetal exposure to PFASs is currently limited to one report on first trimester samples. There is no data available on PFAS concentrations in fetal organs throughout all three trimesters of pregnancy.

Methods: We measured the concentrations of perfluorooctanesulfonic acid (PFOS), perfluorooctanoic acid (PFOA), perfluorononanoic acid (PFNA), perfluorodecanoic acid (PFDA), perfluoroundecanoic acid (PFUnA), and perfluorohexane sulfonic acid (PFHxS) in human embryos and fetuses with corresponding placentas and maternal serum samples derived from elective pregnancy terminations and cases of intrauterine fetal death. A total of 78 embryos and fetuses aged 7–42 gestational weeks were included and a total of 225 fetal organs covering liver, lung, heart, central nervous system (CNS), and adipose tissue were analyzed, together with 71 placentas and 63 maternal serum samples. PFAS concentrations were assayed by liquid chromatography/triple quadrupole mass spectrometry.

Results: All evaluated PFASs were detected and quantified in maternal sera, placentas and embryos/fetuses. In maternal serum samples, PFOS was detected in highest concentrations, followed by PFOA > PFNA > PFDA = PFUnA = PFHxS. Similarly, PFOS was detected in highest concentrations in embryo/fetal tissues, followed by PFOA > PFNA = PFDA = PFUnA. PFHxS was detected in very few fetuses. In general, PFAS concentrations in embryo/fetal tissue (ng/g) were lower than maternal serum (ng/ml) but similar to placenta concentrations. The total PFAS burden (i.e. the sum of all PFASs) was highest in lung tissue in first trimester samples and in liver in second and third trimester samples. The burden was lowest in CNS samples irrespective of fetal age. The placenta:maternal serum ratios of PFOS, PFOA and PFNA increased across gestation suggesting bioaccumulation in the placenta. Further, we observed that the ratios were higher in pregnancies with male fetuses compared to female fetuses.

Conclusions: Human fetuses were intrinsically exposed to a mixture of PFASs throughout gestation. The

Abbreviations: BMI, body mass index; C6, alkane with 6 carbon atoms; C8, alkane with 8 carbon atoms; CNS, central nervous system consisting spinal cord from first trimester and brain from second and third trimesters; IQR, interquartile range; IS, internal standard; IUFD, intrauterine fetal death; IUGR, intrauterine growth restriction; LC/MS/MS, liquid chromatography/triple quadrupole mass spectrometry; LOQ, limit of quantification; n, number of samples; NA, sample not available; NM, sample not measured; PFAS, perfluoroalkyl substance; PFDA, perfluorodecanoic acid; PFHxS, perfluorohexane sulfonic acid; PFNA, perfluorononanoic acid; PFOA, perfluorooctanoic acid; PFOS, perfluorooctanesulfonic acid; PFUnA, perfluoroundecanoic acid; REACH, Registration, Evaluation, Authorization and Restriction of Chemicals; SD, standard deviation; T1, first trimester; T2, second trimester; T3, third trimester

* Corresponding author at: Division of Obstetrics and Gynaecology, Department of Clinical Science, Intervention and Technology, K57 Karolinska University Hospital, Karolinska Institutet, 141 86 Stockholm, Sweden.

E-mail addresses: linn.salto.mamsen@regionh.dk (L.S. Mamsen), richelle.duque.bjorvang@ki.se (R.D. Björvang), daniel.mucs@ki.se (D. Mucs), marie-therese.vinnars@ki.se (M.-T. Vinnars), nikos.papadogiannakis@ki.se (N. Papadogiannakis), christian.lindh@med.lu.se (C.H. Lindh), claus.yding.andersen@regionh.dk (C.Y. Andersen), pauliina.damdimopoulou@ki.se (P. Damdimopoulou).

¹ Contributed equally to this work.

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compounds were detected in all analyzed tissues, suggesting that PFASs reach and may affect many types of organs. Collectively, our results demonstrate that PFASs pass the placenta and deposit to embryo and fetal tissues, calling for risk assessment of gestational exposures.

1. Introduction

The perfluoroalkyl substances (PFASs) are water- and grease-resistant compounds widely used in industrial applications and consumer products (Bach et al., 2015; Johnson et al., 2014). PFASs consist of simple alkyl chains where fluoride atoms replace hydrogen atoms, forming strong carbon-fluorine bonds resistant to degradation. This makes these compounds highly persistent and bioaccumulative in the environment, with longer chains more persistent than shorter ones (Key et al., 1997). Many of the PFASs, such as perfluorooctanoic acid (PFOA), perfluorononanoic acid (PFNA), perfluorodecanoic acid (PFDA) and perfluorohexane sulfonic acid (PFHxS) are listed as Substances of Very High Concern and regulated under the Registration, Evaluation, Authorization and Restriction of Chemicals (REACH) in the EU (REACH, 2017) due to their bioaccumulative properties and toxicity for reproduction (Bach et al., 2015; De Solla et al., 2012; Olsen et al., 2007; Tsuda, 2016). The United States Environmental Protection Agency (2018) also collects data on PFASs in drinking water under the Third Unregulated Contaminant Monitoring Rule. Perfluorooctanesulfonic acid (PFOS) is restricted in accordance to the Stockholm Convention as a persistent organic pollutant (United Nations Environment Programme, 2017). Many short-chain PFASs are still widely used, however, they are believed to be less bioaccumulative and less toxic (Brendel et al., 2018).

PFASs are amphiphilic compounds with estimated octanol-water partition coefficients (logP) varying from 2.82 to 8.90 (Wang et al., 2013), and can be detected in several human organs (Kärman et al., 2010; Maestri et al., 2006; Olsen et al., 2003; Pérez et al., 2013). In general, lung tissue is suggested to accumulate the highest concentration of PFASs, although PFOS predominantly accumulated in liver and PFOA in bones (Kärman et al., 2010; Maestri et al., 2006; Pérez et al., 2013). Mother-to-fetus transport of PFAS has been addressed in one study that measured the concentrations in maternal serum, cord blood and placenta and found that PFOS, PFOA, and PFHxS were efficiently transported from maternal to fetal circulation (Chen et al., 2017).

Prenatal exposure to PFOS and PFOA has been associated with low birth weight in humans (Fei et al., 2007; Johnson et al., 2014; Lauritzen et al., 2017; Maisonet et al., 2012), although reports were inconsistent (Bach et al., 2015; Olsen et al., 2009), underlining the need for more studies. Further, maternal serum concentrations of PFDA and PFNA have been associated with higher miscarriage rates (Jensen et al., 2015), whereas PFOS and PFOA showed limited evidence for an association (Darrow et al., 2014). In rodents, high exposure of PFOS (2–20 mg/kg) and PFOA (1–40 mg/kg) during pregnancy have been associated with reduced postnatal survival, lower birth weight, decreased growth of the pups, disturbed lactation (Lau et al., 2006, 2004, 2003; Olsen et al., 2009), and disrupted thyroid function (Yu et al., 2009). Interestingly, some of the associations appear to be modified by the fetal sex. Concentrations of PFOS and PFOA in cord blood have been associated with an increased risk of congenital cerebral palsy in Danish boys but not in girls (Liew et al., 2014). In contrast, PFOA in cord plasma has been suggested to increase the thyroxine hormone level in newborn girls but not in boys (de Cock et al., 2014). In rodents, prenatal exposure to PFOA advanced the onset of puberty in male mice, with no effect in female mice (Lau et al., 2006). Taken together, data on potential adverse health effects of PFASs on human fetal development are still inconclusive and possible sexually dimorphic associations have not been fully addressed.

Chemical health risk assessment requires data on adverse health effects and chemical exposure levels in the affected population.

Biomonitoring programs have shown that PFASs can be detected in the serum of virtually everyone including pregnant women (Mørck et al., 2015; Starling et al., 2014). However, biomonitoring programs do not contain human fetal samples, and thus there is a lack of data concerning human fetal exposure. We have previously measured tissue concentrations of PFASs in human first trimester embryos and fetuses (gestational weeks 7–12) and found a significant positive correlation between some PFASs and gestational age (Mamsen et al., 2017). The present study expands these analyses by including human fetal tissues from second and third trimester, allowing for an evaluation of human fetal PFASs exposure across the entire gestational period. We report fetal concentrations of six PFASs: PFOS, PFOA, PFNA, PFDA, PFUnA, and PFHxS in a total of 78 embryos and fetuses aged 7–42 gestational weeks. These data provide *i*) important information on the transfer of PFASs from maternal circulation to fetal organs and bioaccumulation during pregnancy; *ii*) actual fetal PFAS concentrations in vital organs during the entire pregnancy, which is essential for better risk assessment and for designing future experimental studies; and *iii*) tools for epidemiologists to estimate fetal exposures based on maternal serum concentrations in pregnancy cohort studies.

2. Materials and methods

2.1. Human embryonic and fetal tissues, placentas and maternal blood samples

First trimester embryonic and fetal tissues, including liver, lung, heart, and spinal cord, together with placenta samples were obtained at the Departments of Obstetrics and Gynaecology, University Hospital Skejby and Regional Hospital Randers, Denmark as described (Mamsen et al., 2017). Tissues were obtained in 2014–2015 from healthy women above the age of 18 years, who had decided to terminate pregnancy for reasons other than fetal abnormality before gestational week 12. All embryos and fetuses appeared morphologically normal. All participants received oral and written information and gave their informed consent. Maternal blood samples were obtained on the day of fetal evacuation. Embryonic and fetal sex were determined by gonadal morphology and confirmed by PCR (Nakahori et al., 1991). PFAS concentrations in first trimester embryonic and fetal tissues has previously been published (Mamsen et al., 2017).

Second and third trimester fetal samples were derived from pregnancies that resulted in intrauterine fetal death (IUFD) which underwent an autopsy to determine cause of death at the Perinatal Pathology Unit at Karolinska University Hospital, Huddinge, Sweden in 2015–2016. The cause of fetal death was determined according to the Stockholm classification (Varli et al., 2008) and was recorded (Supp. Table 1) together with associated clinical data retrieved from autopsy reports and medical records. Fetal sex was determined by gross morphology. Fetal tissues collected included liver, lung, heart, brain, and adipose tissue (when available) together with placenta samples. Additionally, biobanked maternal serum samples obtained during routine first trimester prenatal visit to the maternity care in gestational weeks 9–12 were retrieved from the Stockholm Medical Biobank and included in the analyses. All data were completely anonymized.

2.2. Tissue processing

Tissue samples (mean weight: 0.064 g, range: 0.047–0.139 g) were homogenized at a ratio of three parts 70% acetonitrile solution containing all included isotopically labelled PFASs as internal standards

(IS) to one part tissue on a Tissuelyser II (Qiagen, Copenhagen, Denmark) for 60 s at 15 Hz at room temperature with 2 mm stainless steel beads, thereafter shaken at room temperature for 30 min followed by centrifugation at 1600 × g according to Mamsen et al., 2017.

2.3. Analysis of PFASs

The analyses of PFOS, PFOA, PFNA, PFDA, PFUnA, and PFHxS in serum, placenta and fetal tissue samples were performed by triple quadrupole mass spectrometry (QTRAP 5500; AB Sciex, Foster City, CA, USA) coupled to a liquid chromatography system (LC/MS/MS) (UFLCXR, Shimadzu Corporation, Kyoto, Japan), as described previously (Mamsen et al., 2017). Serum calibration standards have been previously described (Mamsen et al., 2017). Briefly, serum was used as matrix for the calibration standards and also as a proxy matrix for the tissue samples. The samples were prepared using three parts of 70% acetonitrile solution containing the ISs, as above. The PFAS concentrations were quantified by peak area ratios between the analytes and the IS and corrected for the chemical blank. The limit of quantification (LOQ) was defined as the concentration corresponding to ten times the standard deviation (SD) of the ratio of the peak at the same retention time as the analyzed compounds and the corresponding IS divided by the slope of the calibration line and determined in the chemical blank samples (Supp. Table 2). The laboratory is part of the Round Robin inter-comparison program for analyses of PFOS and PFOA (Professor Dr. med. Hans Drexler, Institute and Out-Patient Clinic for Occupational, Social and Environmental Medicine, University of Erlangen-Nuremberg, Germany) with results within the program's tolerance limits.

2.4. Statistical methods

All statistical analyses were performed using the R programming language (R Core Team, 2018) through RStudio (RStudio Team, 2016) with the packages ggplot2 (Wickham, 2009), ggpubr (Kassambara, 2017), reshape2 (Wickham, 2007), gplots (Warnes et al., 2016), and dichromat (Lumley, 2013). Analyses were performed only on samples above LOQ. Kruskal-Wallis test was used to compare chemical concentrations among trimesters and Wilcoxon ranked test with Bonferroni correction for post hoc comparisons. Correlation was investigated using Spearman's rank correlation coefficient. The total PFAS burden for different tissues were calculated by adding the averaged tissue concentrations of the five compounds analyzed in the samples. PFHxS was not included because it was not measured in the first trimester samples. Relative concentrations were computed by dividing the tissue concentration with either the maternal serum or placenta concentration for each compound. The spinal cord of the first trimester and the brain of the second and third trimester fetuses were grouped as central nervous system (CNS) since they are all protected by the blood-brain barrier (Daneman and Prat, 2015). Additionally, a linear regression model was used to determine associations between PFAS tissue concentrations and fetal age in days in both sexes. Significance was defined as $p < 0.05$.

Table 1
Maternal and fetal characteristics by trimester.

Characteristics	First trimester <i>n</i> = 38	Second trimester <i>n</i> = 18	Third trimester <i>n</i> = 22
Maternal age (years)	26.5 ± 6.9 (18–46)	32.5 ± 5.8 (22–45)	32.5 ± 3.7 (27–40)
Gestational age (weeks)	9.5 ± 1.2 (7–12)	24.9 ± 1.5 (22–28)	36.8 ± 3.5 (30–41)
BMI during first visit (kg/m ²)	22.8 ± 4.0 (17.6–32.4)	23.6 ± 2.7 (19.7–28.4)	25.6 ± 4.7 (19.3–35.0)
Fetal sex, male:female, <i>n</i> (%)	21 (61.8): 13 (38.2)	8 (44.4): 10 (55.6)	10 (45.5): 12 (54.5)
Fetal weight (g)	NA	629 ± 138 (435–915)	2839 ± 887 (1050–4180)

Values are *n* (%) or mean ± SD (range). First trimester fetal samples originated from elective abortions, second and third trimester fetal samples originated from intrauterine fetal death.

Abbreviation: BMI – body mass index; NA - Not available.

3. Results

3.1. Sample characteristics

A total of 78 pregnancies were included, from which a total of 296 individual tissue samples were obtained. Maternal blood samples were available in 63 cases. Fetal tissue was obtained from 38 first trimester pregnancies, 18 second trimester pregnancies, and 22 third trimester pregnancies (Table 1). Maternal and fetal characteristics are presented in Table 1. The first trimester women were younger than the second ($p < 0.01$) and third trimester women ($p < 0.001$) (Table 1). Maternal body mass index (BMI) during the first prenatal visit was significantly higher in the third trimester group compared to the first trimester group ($p < 0.05$) (Table 1). The most common causes of death in the IUFD cases were placental insufficiency and intrauterine growth restriction (IUGR) (58%), and infection (13%) with the diagnosis being certain in 60% of the cases (Supp. Table 1).

3.2. PFAS quantification frequency

The LOQ ranged from 0.03 to 0.60 ng/ml for serum samples and from 0.09 to 0.60 ng/g for tissue samples (Supp. Table 2). Samples below LOQ were not included in further analyses. Quantification frequencies of PFASs above LOQ are presented in a heat map to illustrate the typical exposure patterns (Fig. 1). The quantification frequencies were highest in maternal serum samples followed by placenta and fetal tissues. The frequencies increased by trimester in fetal organs. The most commonly detected compound was PFOS followed by PFOA and PFNA. These chemicals were detected above LOQ in 100% of maternal serum samples, in 93%, 82%, and 83% of placenta samples and 76%, 70% and 53% of fetal organs, respectively (Fig. 1, Supp. Table 3). The quantification frequencies of PFDA, PFUnA, and PFHxS in maternal serum samples were 92%, 81%, 65%, respectively. In the placenta, these were quantified at a frequency of 28%, 11% and 0%, respectively, and 15%, 10% and 6% in fetal organs (Fig. 1, Supp. Table 3). In fetal tissues, PFASs were most often detected in liver and lung followed by adipose, heart and CNS (Fig. 1, Supp. Table 3).

3.3. PFASs concentrations in maternal serum, placenta, and fetal tissue

In maternal serum, PFOS was detected in the highest concentration (median 4.91 ng/ml; range 1.04–16.66 ng/ml), followed by PFOA (median 1.55 ng/ml; range 0.55–7.95 ng/ml), and PFNA (median 0.55 ng/ml; range 0.14–2.9 ng/ml) in all trimesters (Fig. 2A, Supp. Table 3). PFDA (median 0.24 ng/ml; range 0.07–1.11 ng/ml), PFUnA (median 0.32 ng/ml; range 0.1–1.73 ng/ml), and PFHxS (median 0.31 ng/ml; range 0.08–5.23 ng/ml) were all detected at similar low concentrations in all trimesters (Fig. 2A, Supp. Table 3). Concentrations of PFOS, PFOA, PFNA, PFDA and PFUnA were significantly positively correlated with each other ($\rho = 0.37$ – 0.83 , $p < 0.05$) except for PFOA with PFUnA (Supp. Fig. 1A). Only PFOS was positively correlated with PFHxS ($\rho = 0.65$, $p < 0.001$) (Supp. Fig. 1A). PFOS and PFNA were

significantly higher in maternal serum from the first trimester compared to the second ($p < 0.001$) and third trimester ($p < 0.001$) (Fig. 2A). The concentration of PFUnA was higher in the first trimester maternal serum samples compared to the third trimester cohort ($p < 0.05$) (Fig. 2A).

In placenta and fetal tissues, highest concentration quantified was PFOS (placenta median 1.24 ng/g, range 0.45–3.87 ng/g; fetal tissues median 0.83, range 0.19–12.61 ng/g) followed by PFOA (placenta median 0.3 ng/g, range 0.15–0.99 ng/g; fetal tissues median 0.49 ng/g, range 0.15–3.62 ng/g) (Fig. 2B, Supp. Table 3). Concentrations of PFNA, PFDA and PFUnA were below 0.73 ng/g in all placenta and fetal tissues analyzed (Fig. 2B, Supp. Table 3). PFHxS was detected only in 6% of all fetal tissues but when present, concentrations were relatively high (median 0.75 ng/g; range 0.62–1.37 ng/g). Generally, the concentrations of PFAS in tissues were positively correlated with maternal serum levels (Supp. Fig. 1B). The correlations reached statistical significance in the case of PFOS in placenta ($\rho = 0.34$, $p < 0.05$), lung ($\rho = 0.34$, $p < 0.05$) and adipose ($\rho = 0.46$, $p < 0.05$) as well as PFOA in placenta ($\rho = 0.32$, $p < 0.05$), liver ($\rho = 0.54$, $p < 0.001$), lung ($\rho = 0.60$, $p < 0.001$) and heart ($\rho = 0.72$, $p < 0.001$) (Supp. Fig. 1B). Moreover, PFNA levels in the lung was also positively correlated with the maternal serum ($\rho = 0.42$, $p < 0.05$) (Supp. Fig. 1B). The placenta and liver concentrations of PFOA were higher in third trimester than in first trimester ($p < 0.01$) (Fig. 2B). Liver concentrations of PFOA and PFNA were higher in second trimester samples than in first trimester cohort ($p < 0.01$), and the liver PFOA concentration in second trimester was also higher than in third trimester ($p < 0.05$) (Fig. 2B). The lung concentration of PFOA in the second trimester was higher compared to the first trimester ($p < 0.05$) (Fig. 2B).

The total concentration of PFASs, or the PFAS burden, was evaluated. PFHxS was not included because it was not measured in first trimester fetal tissues. In the first trimester, the highest burden was found in fetal lung tissue with a total concentration of 3.5 ng/g PFASs (74% PFOS, 9% PFOA, 5% PFNA, 6% PFDA, and 6% PFUnA) (Fig. 3). In second and third trimester, fetal liver had the highest PFAS burden, which in second trimester summed up to 3.4 ng/g (39% PFOS, 34% PFOA, 9% PFNA, 7% PFDA, 11% PFUnA) and in third to 2.9 ng/g (47%

PFOS, 21% PFOA, 8% PFNA, 10% PFDA, 14% PFUnA) (Fig. 3). The CNS samples had the lowest PFAS burden in all trimesters (Fig. 3). Similar results were found in second and third trimesters samples when PFHxS was included (Supp. Fig. 2).

3.4. Maternal characteristics and PFASs concentrations

In our cohort, there was a negative correlation between BMI and maternal serum concentrations of PFNA ($\rho = -0.26$, $p < 0.05$) and PFUnA ($\rho = -0.3$, $p < 0.05$). However, this correlation was not found in the placenta and fetal organ concentrations. Maternal age correlated positively with the following compounds: PFOS in fetal adipose ($\rho = 0.55$, $p < 0.05$), PFOA in fetal adipose ($\rho = 0.44$, $p < 0.05$), and PFDA in placenta ($\rho = 0.49$, $p < 0.05$) and fetal liver ($\rho = 0.52$, $p < 0.05$). No other significant correlations between PFAS concentrations, maternal age and BMI were found.

3.5. PFASs in human fetal organs in relation to maternal concentrations

Maternal serum PFAS concentrations varied between trimesters and this could affect fetal tissue concentrations and evaluation of fetal transfer. We therefore normalized fetal tissue concentrations (ng/g) by maternal serum concentrations (ng/ml). These fetal tissue:maternal serum ratios were calculated only in those cases where data ($> \text{LOQ}$) for paired samples (i.e. maternal serum and corresponding placenta and fetal tissues) were available, and where at least three independent values (i.e. individual fetuses) could be calculated. The tissue:maternal serum ratio presented in percent showed that tissues had, in general, lower exposure than serum (median ratios $< 100\%$) (Fig. 4A). For PFOS, the highest relative ratios were found in placenta and liver, and for PFOA and PFNA, in liver and adipose (Fig. 4A). The lowest fetal tissue:maternal serum ratios for all compounds were found in CNS samples (Fig. 4A). The median ratios were in general higher in second and third trimester compared to first trimester. The ratios of PFOS, PFOA, and PFNA in placenta and liver were significantly higher in the second and third trimesters compared to the first ($p < 0.01$) (Fig. 4A). The fetal lung:maternal serum ratio of PFOA and PFNA were also

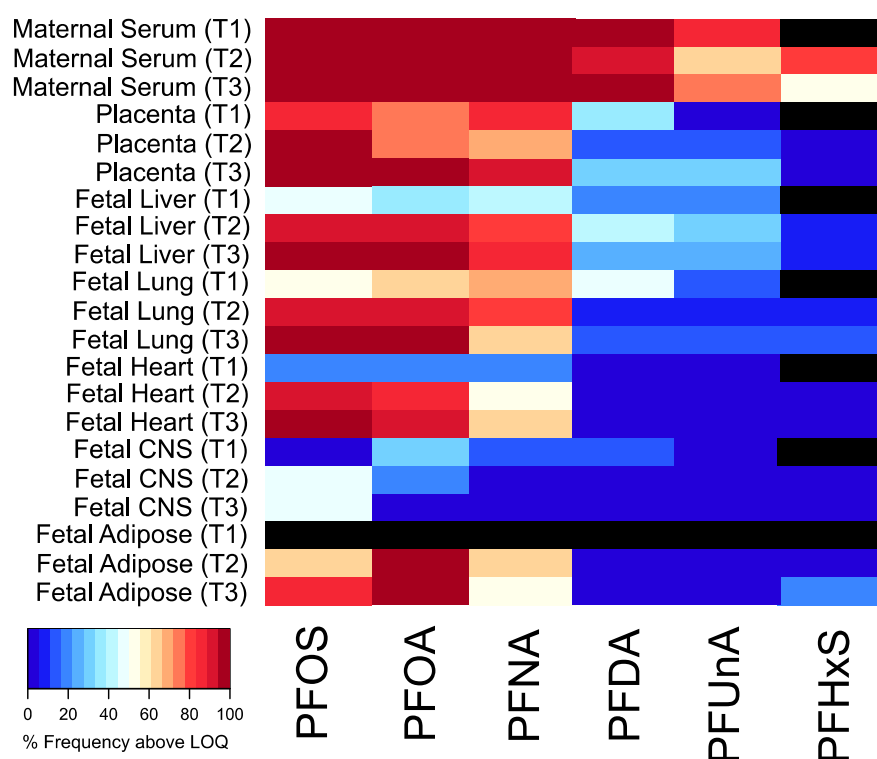


Fig. 1. Quantification frequency of perfluorinated alkyl substances (PFAS) in maternal serum, placenta, and fetal organs by trimester.

Heat map illustrating the quantification frequency of PFASs in maternal serum, placenta and fetal organs according to trimester (T1, T2, T3). Frequency varies from PFAS quantified in 0–5% of samples (dark blue) to PFAS quantified in 95–100% samples (dark red). Black color indicates samples that were not available or chemical that was not measured. Number of observations varies and can be found in Supp. Table 3. Abbreviations: PFOS – perfluorooctanesulfonic acid; PFOA – perfluorooctanoic acid; PFNA – perfluorononanoic acid; PFDA – perfluorodecanoic acid; PFUnA – perfluoroundecanoic acid; PFHxS – perfluorohexane sulfonic acid; CNS – central nervous system consisting spinal cord from first trimester and brain from second and third trimesters.

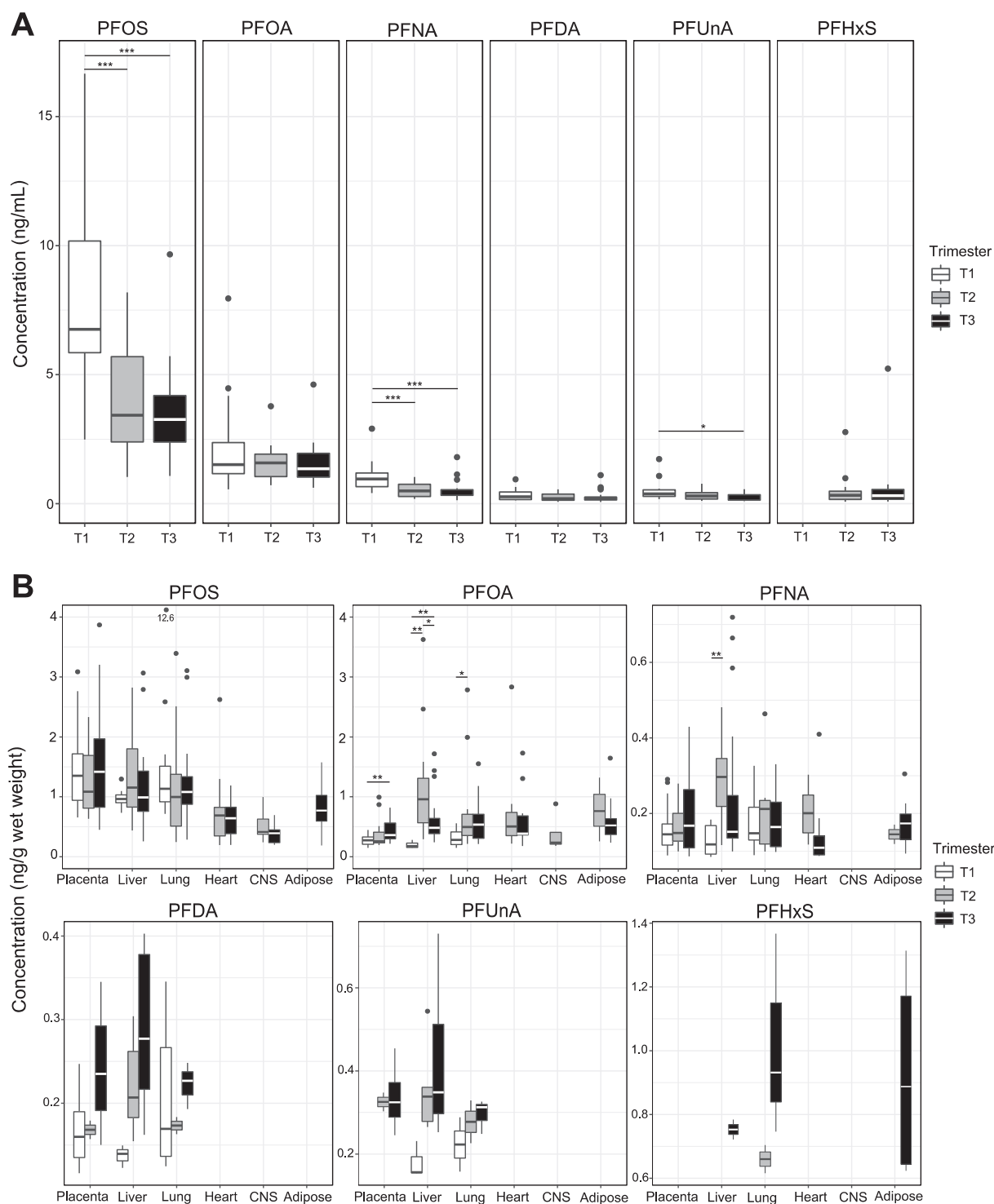


Fig. 2. Concentration of perfluorinated alkyl substances (PFAS) in maternal serum, placenta and fetal tissues.

(A) PFAS concentrations in maternal serum (ng/mL). All maternal serum samples were collected during the first trimester and the data are presented per trimester according to time of fetal death. (B) PFAS concentration in fetal tissue samples (ng/g). The data are presented as boxplots where line is median, the box is the interquartile range (IQR), and whiskers are $1.5 \times \text{IQR}$. Outliers are defined as values $> 1.5 \times \text{IQR}$ and depicted with dots. In case of extreme outliers, the value is shown next to the dot. Numbers of observations varies and can be found in Supp. Table 3. PFHxS was not measured in first trimester samples. Statistical differences between trimester were tested with Wilcoxon rank-sum test with Bonferroni correction and shown as $*p < 0.05$; $**p < 0.01$; $***p < 0.001$. Abbreviations: PFOS – perfluorooctanesulfonic acid; PFOA – perfluorooctanoic acid; PFNA – perfluorononanoic acid; PFDA – perfluorodecanoic acid; PFUnA – perfluoroundecanoic acid; PFHxS – perfluorohexane sulfonic acid; CNS – central nervous system consisting spinal cord from first trimester and brain from second and third trimesters; T1 – first trimester; T2 – second trimester; T3 – third trimester.

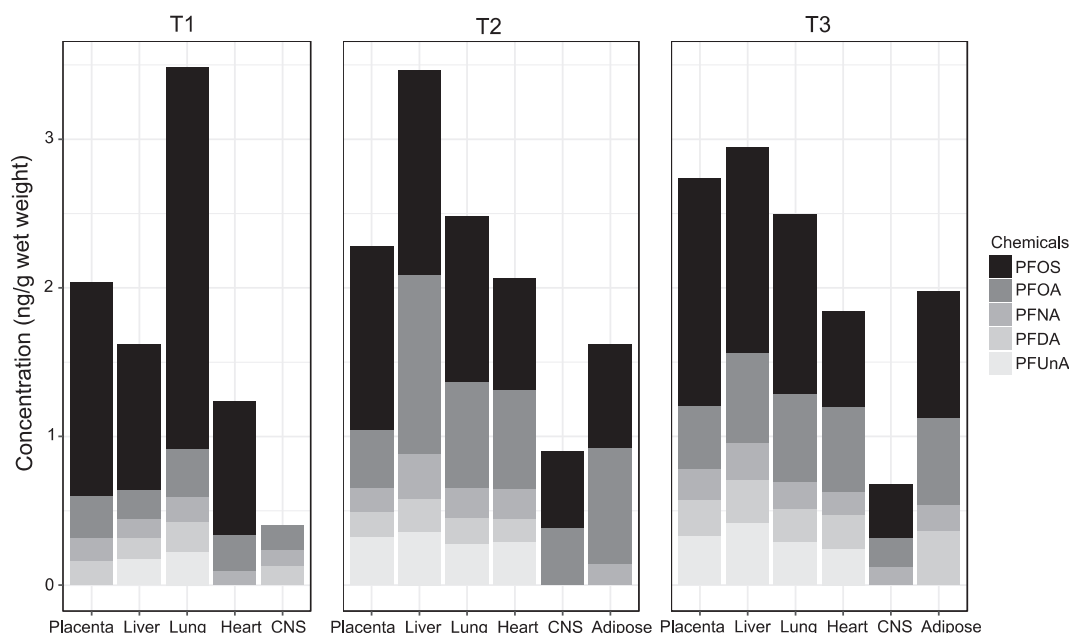


Fig. 3. Total burden of perfluorinated alkyl substances (PFAS) per organ.

The total PFAS burden (ng/g) in the samples was calculated by adding the averaged sample concentrations of the five analyzed compounds. PFHxS was not included because it was not measured in the T1 fetal tissues. There was no adipose tissue in T1 fetuses. Abbreviations: PFOS – perfluorooctanesulfonic acid; PFOA – perfluorooctanoic acid; PFNA – perfluorononanoic acid; PFDA – perfluorodecanoic acid; PFUnA – perfluoroundecanoic acid; PFHxS – perfluorohexane sulfonic acid; CNS – central nervous system consisting spinal cord from first trimester and brain from second and third trimesters; T1 – first trimester; T2 – second trimester; T3 – third trimester.

significantly higher in the second and third trimesters compared to the first trimester ($p < 0.05$).

All maternal serum samples were obtained during first trimester while tissue samples for second and third trimesters were collected after fetal death. We therefore also calculated fetal exposures in relation to placenta exposure, as placenta and fetal tissues represent the same sample collection time point. The fetal tissue:placenta ratios are presented in Fig. 4B. In contrast to the tissue:serum ratios that remained $< 100\%$, the tissue:placenta percentages were clearly higher (Fig. 4B). The tissue:placenta ratios of PFOA, PFNA, PFDA, and PFUnA were close to or above 100% indicating similar or higher exposure in fetal organs compared to placenta (Fig. 4B). Interestingly, several tissue:placenta ratios of PFOA and PFNA were highest in second trimester samples compared to the first and third trimester. The highest fetal tissue:placenta ratios were detected in liver and the lowest in CNS.

3.6. Accumulation of PFASs during gestation

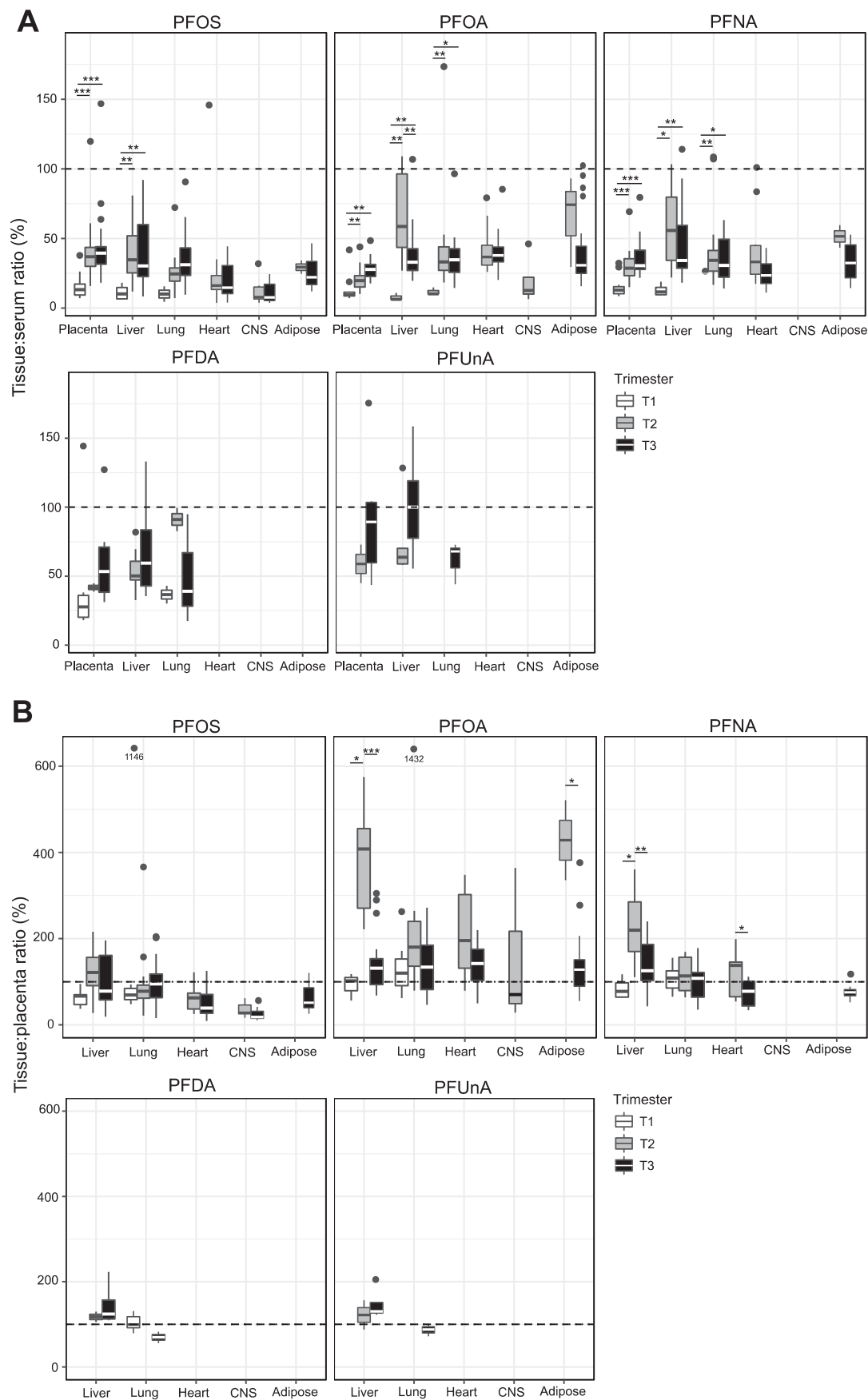
Possible bioaccumulation of PFASs in fetal tissues across gestation in the whole cohort and in female and male fetuses separately was evaluated. Due to the variation in maternal serum concentrations (Fig. 2A), fetal tissue concentrations were normalized to maternal serum before studying associations between fetal age and PFASs concentrations in tissues. Accumulation in placenta and fetal tissues across gestation was evaluated by plotting the normalized concentrations as a function of fetal age in days. Evidence of bioaccumulation was detected in placenta for the following compounds: PFOS ($p < 0.01$), PFOA ($p < 0.001$), and PFNA ($p < 0.001$) (Fig. 5). The placenta:serum ratio (%) of PFOS, PFOA, and PFNA increased by 0.13% ($p < 0.01$), 0.07% ($p < 0.001$), and 0.11% ($p < 0.001$), respectively, for every day increase in fetal age. Interestingly, the placenta:serum ratio of PFOA was 5.6% higher in male fetuses than in female fetuses ($p < 0.05$) (Fig. 5). The placenta:serum ratios of PFOS and PFNA were also higher in male fetuses than in female fetuses but the difference did not reach statistical significance (Fig. 5). We also studied bioaccumulation in fetal tissues but did not find statistically significant associations between

normalized exposure levels and fetal age.

4. Discussion

4.1. PFASs in fetal tissues and maternal serum

The concentrations of six PFASs were analyzed in a total of 225 human fetal organs, which to our knowledge is the largest human fetal cohort available covering all three trimesters of pregnancy. Similar studies are difficult to carry out in experimental settings using typical laboratory animals such as mice and rats due to the very small size of their fetal organs and technical limitations in detection limits. The reliability of rodents as models for human PFAS exposure could also be questioned due to faster elimination of PFAS compounds in rats compared to humans (Chang et al., 2012; Olsen et al., 2007; Sundström et al., 2012; Vanden Heuvel et al., 1991). The maternal serum PFAS concentrations measured in our cohort were similar to recent reports, with the highest PFAS concentrations being PFOS, PFOA and PFNA (Calafat et al., 2007; Manzano-Salgado et al., 2015; Mørck et al., 2015; U.S. Department of Health and Human Services, 2018). It is interesting to note that the maternal serum PFOS and PFNA concentrations were higher in the first trimester cohort compared to the second and third trimester cohorts. Because all maternal serum samples were taken during the first trimester, this difference may be related to differences between the cohorts where the first trimester samples were obtained from Denmark while the second and third trimesters were from Sweden. However, PFAS levels in Danish and Swedish pregnant women are generally similar (Jensen et al., 2015; Mamsen et al., 2017; Ode et al., 2013). Year of sampling could also affect the levels, but in our study, all samples were collected within a short period of time (2014–2016). Due to the differing maternal serum levels, we have analyzed the fetal exposure data both as absolute concentrations and as relative levels normalized to maternal serum. In fetal tissues, PFOS, PFOA and PFNA were detected in 76%, 70%, and 53% of the organs, respectively, whereas PFDA, PFUnA, and PFHxS were detected only in 15%, 10%, and 6% of the organs, respectively, suggesting that both



(caption on next page)

Fig. 4. Perfluorinated alkyl substances (PFASs) in fetal organs in relation to maternal concentrations.

(A) Fetal tissue:maternal serum ratios of PFASs were calculated by dividing the fetal tissue concentration by the maternal serum concentration. (B) Fetal tissue:placenta ratios of PFASs were calculated by dividing the fetal tissue concentration by the placenta concentration. Dotted lines marks 100% at which the concentrations in fetal organs equals (A) maternal serum or (B) placenta. Boxplots consist of the interquartile range (IQR) and the median, and whiskers are $1.5 \times \text{IQR}$. Outliers are defined as values $> 1.5 \times \text{IQR}$ and depicted with dots. In case of extreme outliers, the value is written next to the dot. Number of observations varies and can be found in Supp. Table 3. Statistical significance between trimesters was tested using Wilcoxon rank-sum test with Bonferroni correction and is marked $*p < 0.05$; $**p < 0.01$; $***p < 0.001$. Abbreviations: PFOS – perfluorooctanesulfonic acid; PFOA – perfluorooctanoic acid; PFNA – perfluorononanoic acid; PFDA – perfluorodecanoic acid; PFUnA – perfluoroundecanoic acid; PFHxS – perfluorohexane sulfonic acid; CNS – central nervous system consisting spinal cord from first trimester and brain from second and third trimesters; T1 – first trimester; T2 – second trimester, T3 – third trimester.

long ($\geq \text{C8}$ -based) and short chained (C6-based, i.e. PFHxS) PFASs are transferred from mother to fetus, although with different efficiencies.

The highest fetal PFAS burden was detected in liver and lung, suggesting that these highly vascularized tissues are prone to accumulate PFASs to a higher extent than other organs. In adults, lung tissue has been found to accumulate the highest concentration of PFASs in an evaluation of brain, liver, lung, bone and kidney samples (Pérez et al., 2013), which the present data supports and extends. In second trimester, the highest concentrations of PFOA and PFNA were detected in the liver, which were higher than first and third trimester samples. This could be explained by the architecture of the prenatal hepatic vascular system where a substantial part of the oxygenated umbilical venous blood bypasses the liver via *ductus venosus* and is delivered directly to the heart from which it is distributed to the whole fetus (Collardeau-Frachon and Scoazec, 2008). The fraction of blood flow shunted from the liver decreases through gestation with a steep fall from 60% to 30% in week 20–26, resulting in an increased umbilical venous flow through the liver (Pennati et al., 2003). This may explain why PFASs accumulate in the liver exactly during this developmental period. Interestingly, CNS was less exposed to all evaluated PFASs than the rest of the tissues, which may suggest that the CNS is protected from exposure by the blood-brain barrier already during fetal development. Compared to reported adult tissue PFAS levels, the concentrations measured in our fetal cohort remained lower except for PFOA in liver, where levels were similar to adults, and for PFDA, PFUnA and PFHxS in liver, where our fetal cohort had higher concentrations than reported for adults (Kärman et al., 2010; Pérez et al., 2013).

4.2. Fetal age positively associated with normalized placental concentrations of PFOS, PFOA and PFNA

Fetal age was positively associated with the placenta:serum ratios of the three most prevalently detected substances (PFOS, PFOA and

PFNA), suggesting that these substances accumulate in the placenta across gestation and give rise to an increased fetal burden. These observations are in line with the well-known properties of PFASs as compounds that are persistent and bioaccumulative. Previous studies have shown that PFOS accumulates in human amniotic fluid during pregnancy (Jensen et al., 2012), supporting the present findings that PFASs may accumulate in human compartments across gestation. We did not find other significant associations between fetal age and normalized tissue exposure levels. This could partly be explained by the lower quantification frequency of the compounds in fetal tissues as compared to placenta, leading to a lower number of paired samples (i.e. maternal serum and tissue) available for analysis and thereby, lower statistical power. Moreover, as discussed above, some PFASs were present in the highest concentration during the second trimester. For example, both absolute and normalized levels of PFOA and PFNA in liver were the highest in the second trimester compared to the first and third. This creates a non-linear exposure pattern, which would not show significant association with fetal age in linear regression modelling.

Interestingly, the accumulation of PFAS in placenta was higher in pregnancies with male fetuses compared to pregnancies with female fetuses, suggesting that PFASs may have a sexually dimorphic accumulation pattern. Higher PFAS serum concentrations in men than women have been described in adults (Góralczyk et al., 2015; Nakayama et al., 2005), and the present data suggest that this dimorphic accumulation may already be found during fetal life. Interestingly, sexually dimorphic adverse outcomes of PFAS exposure have been described in the literature. For example, in rodents, prenatal exposure to PFOA had a significant effect on the fetal growth and development of males, but to a much lesser extent in females (Nakayama et al., 2005; Negri et al., 2017). Potentially differing intrinsic exposure levels between sexes could contribute to this phenomenon. The higher sensitivity of male fetuses to PFOA may be related to a less efficient elimination in males, which have also been seen in rodents (Vanden

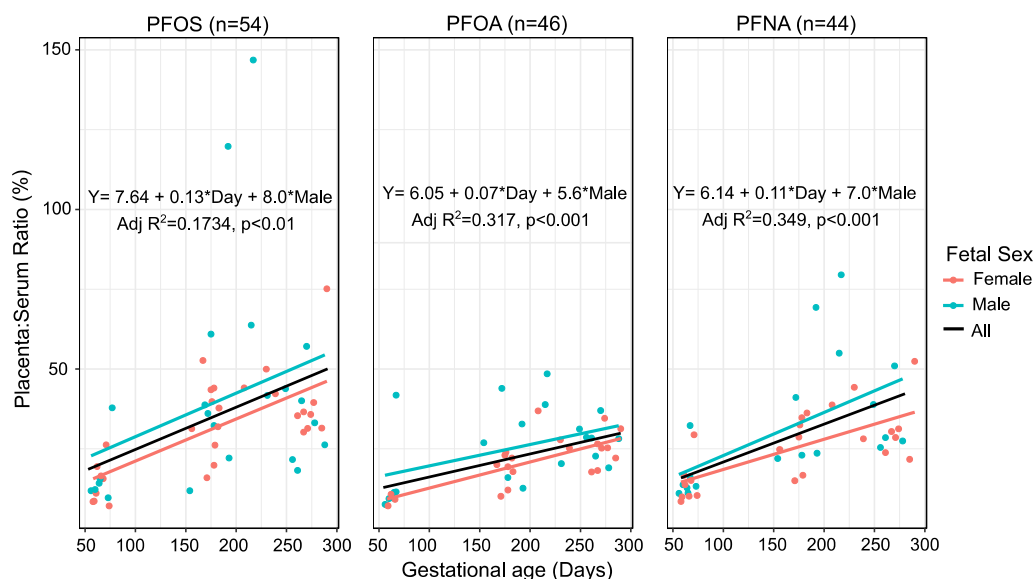


Fig. 5. Effect of gestational age in days and fetal sex on levels of perfluorinated alkyl substances (PFASs) levels in placenta.

Placenta:serum ratios (%) were calculated by dividing placenta concentrations by maternal serum concentrations and plotted as a function of fetal age (days) for male and female fetuses. Linear regression was used to study the significance of the associations as well as differences between sexes. Abbreviations: PFOS – perfluorooctanesulfonic acid; PFOA – perfluorooctanoic acid; PFNA – perfluorononanoic acid.

Heuvel et al., 1991). In humans, prenatal exposure to endocrine disrupting chemicals, including PFASs, has been associated with small-for-gestational-age birth weights particularly in boys, suggesting that males may be more susceptible to chemical exposures than females (Lauritzen et al., 2017). Moreover, there are differences in placental function between male and female fetuses, which could lead to differing cumulative exposure over gestation (Rosenfeld, 2015). Higher placental vascular resistance has been found in pregnancies with female fetuses compared to pregnancies with male fetuses (Widnes et al., 2017), which could affect blood flow to the fetus and create different chemical exposure between sexes.

4.3. Maternal BMI, age and PFASs concentrations

The present data suggest a negative correlation between BMI and serum PFNA and PFUnA concentrations, which supports previous findings (Kato et al., 2014; Lauritzen et al., 2016; Mamsen et al., 2017). However, data on BMI and PFAS concentrations are inconsistent with reports available on both no correlations (Nelson et al., 2010) or positive correlations (Fei et al., 2007). In addition, higher concentrations of PFOS and PFOA were observed in cord blood of both underweight and overweight women (Apelberg et al., 2007). These data, together with previous findings, suggest the impact of BMI on PFAS accumulation in women may be modified by multiple yet unknown confounding factors. Maternal PFAS concentrations have previously been suggested to increase with age (Kato et al., 2014; Lien et al., 2013; Mørck et al., 2015). The present data can confirm that maternal age may be a risk factor for increased fetal PFAS exposure burden.

4.4. Maternal PFAS concentrations are reflected in the fetus

In maternal serum, the concentrations of PFASs were positively correlated with each other. These correlations indicate that an exposure pattern exist, where higher exposure to one PFAS is connected to higher exposure to other PFASs, confirming previous findings (Mørck et al., 2015). Further, levels of PFOS, PFOA and PFNA in the maternal sera were positively correlated with those in placenta and corresponding fetal tissues, indicating that maternal PFASs exposure were mirrored in the fetus and suggesting that a high maternal PFASs burden translate to a high fetal burden. Correlated exposure patterns has previously been detected in maternal serum and umbilical cord serum (Chen et al., 2017; Kato et al., 2014; Manzano-Salgado et al., 2015).

4.5. Limitations

Second and third trimester fetal tissues were obtained from IUFD cases, which deviate from normal pregnancies as the fetus died in utero and may not be representative of normal fetuses. However, the IUFD cases in our cohort were well-characterized, and the main causes of death were placental insufficiency/IUGR and infections, which may have limited impact on PFAS concentrations as there were no significant differences on PFAS levels found between IUGR and non-IUGR cases (data not shown). Moreover, geographical differences may play a role in exposure variation as the first trimester came from Denmark and the second and third trimesters came from Sweden. While serum PFAS concentrations have previously been measured in Danish and Swedish pregnant women and are generally similar (Jensen et al., 2015; Mamsen et al., 2017; Ode et al., 2013), we addressed these differences in exposure by normalizing the tissue levels with the concentrations in the maternal serum.

5. Conclusions

Here we report human fetal tissue exposure of PFASs across the entire gestation, together with concentrations in placental and maternal serum. These data are valuable for risk assessors, experimental

biologists, epidemiologists and ecotoxicologists working with PFASs. The values presented may even allow extrapolation of fetal exposure of other similar chemical compounds through in silico modelling. All evaluated PFASs crossed the placenta and reached the fetal organs throughout gestation, although with different penetration efficiencies. PFOS, PFOA, and PFNA were detected in the highest concentrations both in maternal serum, placenta, and in fetal organs. Further, these compounds accumulated in placenta across gestation, suggesting that exposure increases with fetal age. The highest combined PFAS exposure in fetal organs was found in lung and liver, and the lowest in CNS samples. Only six PFASs were included in this study and considering that the PFAS family contains hundreds of congeners, the total PFAS burden can, therefore, be expected to be higher. These observations call for urgent action to better map the adverse health effects of PFAS during fetal development to revise safe exposure levels in humans.

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.envint.2019.01.010>.

Ethical approval

This project was approved by the Research Ethics Committees of the Regional Capital (H-KF 01258206) (first trimester), and the Regional Ethical Review Board in Stockholm (EPN 2015/796-31/2) (second and third trimesters).

Authors' roles

LSM drafted the paper, collected the human first trimester tissues, prepared samples for chemical analysis, and interpreted the data. RDB drafted the paper, did the statistical analyses and prepared figures and tables. DM assisted with data interpretation and English proofreading of the manuscript. M-TV and NP collected the IUFD fetal samples and carried out clinical data collection and cause of death investigations. CHL was responsible for the design of the chemical analysis and interpreted the data. CYA interpreted the data and was responsible for the study design. PD interpreted data, drafted the paper and was responsible for the study design. All authors approved the last version of the manuscript.

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Financial interest declaration

The financial supporters had no role in the study design, collection and analysis of data, data interpretation or in writing the report. The corresponding author had full access to all data presented in the study and had the final responsibility for the decision to submit this paper for publication.

Declarations of interest

None.

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